LECTURE ACTIVITY 3: CARE FOR A SAMPLE? – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is based on one created by the STAT 1110 Course Coordination Team, with thanks

Learning Objectives:

By the end of this activity, you should be able to

1) Describe and identify the population of interest.
2) Collect and identify a sample.
3) Identify the parameter of interest.
4) Identify and calculate the statistic of interest.
5) Identify biased and unbiased sampling methods based on the distribution of statistics.
6) Identify tendency in bias (overestimate or underestimate the parameter of interest).
7) Collect a simple random sample.
8) Explain why increasing the sample size does not make a sample more representative of the population.

Key Terms:

- Population
- Parameter
- Sampling Frame
- Sample Statistic
- Convenience Sample
- Simple Random Sample

Introduction

The **Gettysburg Address** is a fairly famous speech President Abraham Lincoln gave on November 19, 1863, on the battlefield near Gettysburg, PA:

> *Four score and seven years ago our fathers brought forth on this continent, a new nation, conceived in Liberty, and dedicated to the proposition that all men are created equal.*
>
> *Now we are engaged in a great civil war, testing whether that nation, or any nation so conceived and so dedicated, can long endure. We are met on a great battle-field of that war. We have come to dedicate a portion of that field, as a final resting place for those who here gave their lives that that nation might live. It is altogether fitting and proper that we should do this.*
>
> *But, in a larger sense, we can not dedicate – we can not consecrate – we can not hallow – this ground. The brave men, living and dead, who struggled here, have consecrated it, far above our poor power to add or detract. The world will little note, nor long remember what we say here, but it can never forget what they did here. It is for us the living, rather, to be dedicated here to the unfinished work which they who fought here have thus far so nobly advanced. It is rather for us to be here dedicated to the great task remaining before us – that from these honored dead we take increased devotion to that cause for which they gave the last full measure of devotion – that we here highly resolve that these dead shall not have died in vain – that this nation, under God, shall have a new birth of freedom – and that government of the people, by the people, for the people, shall not perish from the earth.*[1]

[1] https://www.abrahamlincolnonline.org/lincoln/speeches/gettysburg.htm

The main question we will work toward answering today is: what is the average (mean) length of words in the Gettysburg Address?

Pro Tip

For this course, unless otherwise specified, "average" will mean the same thing as "mean."

Of course, we could try and calculate this directly - but, where's the fun in that? Rather, let's see if we can approach this in a more statistical fashion.

Terminology

Recall the following terms you learned about from the textbook:

- The **population** refers to the set of observational units of interest
  - Note: a "population" in the statistical sense doesn't necessarily refer to people. Remember (from our first lecture) that observational units can be pretty much anything (cats, cars, etc.)
- A **sample** is a subset of the population (i.e. a collection of units from the population)
  - The number of units included in a sample is called the **sample size**. If a specific number isn't given, we'll often use the letter n to denote the sample size.

Question 1
Describe an observational unit, in the context of this dataset.
Solution: One observational unit is **one word**.

Question 2
Describe the population, in the context of this dataset.
Solution: The population is **the set of all words in the Gettysburg Address**.

A sample, in the context of this problem, is therefore just a collection of words taken from the Gettysburg Address. Let's take a look at one particular sample:

{Liberty, nation, war, brave, dedicated, government, perish, devotion, people, remember }

Question 3
Fill In the Blank: This sample has a sample size equal to ___10___. (The answer here is not just n!)
Solution: Since each observational unit is a word, a sample is then a collection of words. The sample size is then just the number of words in the sample.

Because we're ultimately interested in the average length of all words in the Gettysburg Address, let's record the lengths of the words in this sample:

$$\{ 7, 6, 3, 5, 9, 10, 6, 8, 6, 8 \}$$

This list represents a set of recorded values of a variable which we can call **Word Length**, which tracks the lengths of words.

> **Question 4**
> Is the **Word Length** variable categorical or quantitative?
> ◯ Categorical   √ **Quantitative**

We can examine the distribution of sampled word lengths using a dotplot:

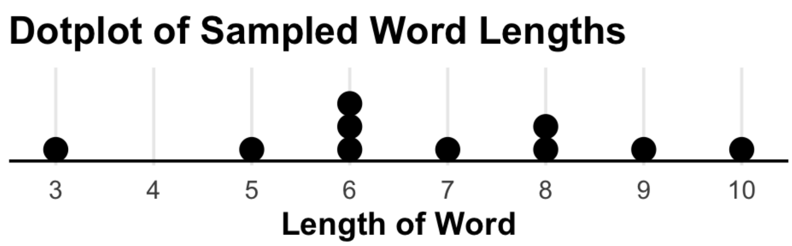


Figure 1: Dotplot of the 10 word lengths

> **Question 5**
> What does each dot in Figure (1) represent?
> **Solution:** Each dot represents one of the 10 sampled words and its length.

> **Pro Tip**
>
> This (hopefully) highlights one key difference between histograms and dotplot: dotplots more easily (from a visual standpoint) display the individual observational units. Though information on observational units is still included in a histogram, the act of representing a unit as a dot is sometimes more appealing.

Statistical Inference

Recall, from the reading, that **statistical inference**, broadly speaking, refers to the act of using samples to draw conclusions about the population from which the sample was taken. A **population parameter** (often shortened to just **parameter**) is a number that describes some aspect of the population. Examples include:

- A population mean, which we denote by μ (the Greek letter "mu")
- A population proportion, which we denote by π (the Greek letter "pi")
- A population standard deviation, which we denote by σ (the Greek letter "sigma")

A **sample statistic** (often shortened to just **statistic**) is a number that describes some aspect of a sample. Examples include:

- A sample mean, which we denote by $\bar{x}$ (pronounced "x-bar")
- A sample proportion, which we denote by $\hat{\pi}$ or $\hat{p}$ (pronounced "pi-hat" or "p-hat")

- A sample standard deviation, which we denote by s

⚠ Caution

Eventually you will be responsible for recognizing the correct notation for each of these! For now we'll be a little lax about that, though.

Pro Tip

If it helps, you can think of statistics as "sample analogs" of population parameters. That is: every population parameter has a corresponding sample statistic. A potentially helpful mnemonic: Sample Statistics Estimate Population Parameters.

Question 6

In the context of the Gettysburg Address:

(a) Is "the average length of all words in the Gettysburg Address" a parameter or a statistic? Why?

√ **Parameter** ◯ Statistic

Solution: This is a quantity that describes the entire population; hence, it is a parameter.

(b) Is "the average length of a sample of 10 words taken from the Gettysburg Address" a parameter or a statistic? Why?

◯ Parameter √ **Statistic**

Solution: This is a quantity that describes the a sample from the population; hence, it is a statistic.

Sampling

For the rest of today's activity, we'll explore different sampling methods to generate samples from the Gettysburg Address. We use the term **sampling method** (aka **sampling scheme** or **sampling technique**) to refer to how a given sample was taken. Two of the sampling methods you saw in the reading are:

- **Convenience Sample**: we simply pick our sample based on what is most convenient
- **Simple Random Sample**: we ensure every sample of size n is equally likely to have been chosen

We use the term **sampling frame** to refer to the set of all units that could theoretically be included in a sample, given a particular sampling method.

To illustrate the notion of a sampling frame, consider the following scenario: suppose we volunteer for a local congressperson and want to estimate the proportion of San Luis Obispo residents that will vote for this congressperson in an upcoming election. Volunteers often address this by sending a text survey to eligible voters' smartphones, asking them questions like "how likely are you to vote for the congressperson in the upcoming election?"

The population, in this case, is the set of all eligible San Luis Obispo voters. But is the sampling frame the entire population - that is, could *every* eligible voter theoretically be included in our sample? The answer is, no - what about eligible voters *that don't have a smartphone*? Given this sampling method (reaching a handful of people by text), not everyone can be reached; the sampling frame is the set of San Luis Obispo residents who have a smartphone, and is therefore smaller than the entire population. Keep this in mind for the next section of this Activity.

First Sampling Method: Convenience Sampling

Let's start by taking a convenience sample, where we pick any 10 words we like.

Question 7
What is the sampling frame using this technique? Specifically, is it larger than, smaller than, or the same as the population?

Solution: The sampling frame is the same as the population, since we could theoretically pick any word from the Address to be in our sample.

Question 8
Pick 10 words from the Gettysburg address. Write them down below, and the corresponding number of letters in each; then calculate the average (mean) length of your selected 10 words. **If you're working in groups, please have each group member complete this independently. Answers will almost certainly vary!**

Word	Length
Four	4
note	4
living	6
devotion	8
met	3

Word	Length
are	3
we	2
highly	6
here	4
consecrate	10

Average length of your 10 words: 5

A notion closely related to that of generalizability is the **bias** of a sampling method. Broadly speaking, we say that a sampling scheme is biased if the sample statistics it produces tend to over- or under-estimate the true value of the population parameter. Because bias involves comparing against the true value of the parameter (a value that is, in practice, often unknown) it can be a little challenging to assess whether a sampling technique is biased or not. Typically, we need detailed information on how the sample was collected, and then we'll need to make a determination based on that information whether we believe the technique to be biased or not.

For example, it is a documented phenomenon that, when scanning a paragraph, people's eyes tend to be drawn toward longer words (as opposed to shorter words).

Question 9

Based on this fact, do you think our sampling technique (asking you to just pick any 10 words you like) is biased or not? If it is biased, do you think it will tend to **overestimate** (consistently produce statistics higher than the true value of the parameter) or **underestimate** (consistently produce statistics lower than the true value of the parameter)?

Solution: Given that people tend to pick longer words, we suspect this sampling technique will be biased and consistently overestimate the true value of the parameter of interest. That is, people will tend to pick longer words than are typical of the Gettysburg Address, leading to an overinflated sense of what the average length of words truly is.

In this particular case, we can "cheat" a little bit - that's because we actually can compute the value of the population parameter directly.

⚠ Caution

Remember - in general we won't know the true value of the population parameter!

It turns out that the average (mean) length of all words in the Gettysburg Address is 4.24 characters. This fact enables us to check the bias of our sampling technique in the following way:

1) Generate a bunch of samples using our technique
2) Calculate the appropriate sample statistic from each
3) Create a dotplot of these sample statistic values, and check how they compare to the true value of the parameter (which, in this case, is 4.24)

I went ahead and did this - I "found[2]" 64 different students and asked them each to select 10 words from the Gettysburg Address, calculate the average length of their sampled words, and report that average back to me. Using these 64 measurements - which had an average of 5.42 - I generated the following dotplot:

[2] Actually, these are fictional students - I used a computer to do this

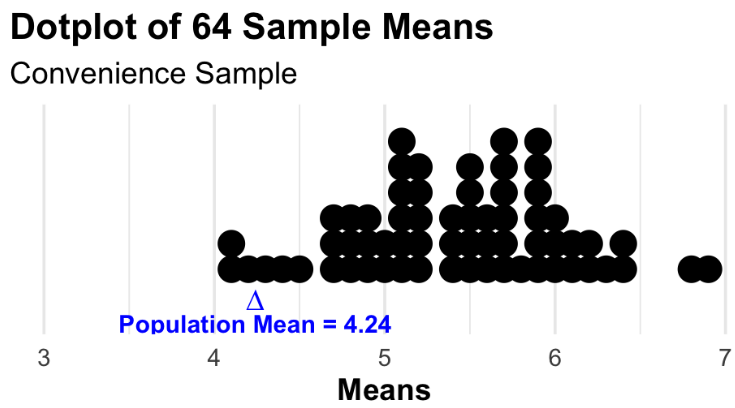


Figure 2: Dotplot of the 64 sample mean word lengths, selected by the (hypothetical) students.

Question 10

What does a dot on this dotplot represent? Is each dot a parameter or a statistic value?

Solution: Each dot represents the average length of a student's sample of 10 words from the Gettysburg Address, and therefore represents a statistic value.

Question 11

Based on the dotplot above, as well as the value of the population parameter (indicated on the dotplot), do you think this sampling scheme (asking you to pick out 10 words on your own) is biased or unbiased? If it is biased, in which direction - do you think there is a tendency to overestimate or underestimate? What about the plot queues you in to this?

Solution: Our suspicions from Question (9) above are confirmed; this technique tends to overestimate the true population mean. This can be seen from the fact that the points on the dotplot fall disproportionately to the right of the true value (of 4.24).

Sample Size and Bias

It's a somewhat common misconception that increasing the sample size can correct for bias. Bias (as it's been discussed in today's lecture activity) is something that describes an entire *sampling method* - as such, increasing the sample size doesn't automatically fix bias!

As an example, suppose we're trying to estimate the true average monthly income of all US residents. Consider the following sampling method: we only visit the *Rosewood* Miramar Beach (an ultra-luxury resort just outside of Santa Barbara, where the cheapest room typically runs for over $1,000 a night), randomly select 10 people, and ask them their monthly income.

Question 12

(a) Describe an observational unit in the context of this problem.

Solution: An observational unit is one person.

(b) What is the population of interest?

Solution: The population is the set of all US residents.

(c) What is the sampling frame?

Solution: The sampling frame is the set of all people staying at the *Rosewood* Miramar Beach.

Naturally, we would expect this sampling method to be biased with a tendency to overestimate - people staying at the *Rosewood* Miramar Beach are likely to be quite affluent, and therefore have higher-than-average incomes. Increasing the sample size from 10 to, say, 100 still wouldn't fix this if we still only sample from the *Rosewood* Miramar Beach!

Second Sampling Method: Simple Random Sampling

Let's turn our attention to simple random sampling! Recall that a simple random sample is one in which every sample of size n is equally likely to have been chosen. (Equivalently, each unit in the population has the same chance of being included in the sample.)

Here's one way we can obtain a simple random sample of size 10 words from the Gettysburg Address:

1) Use a Random Number Generator to randomly pick 10 numbers from the set {1, 2, 3, ..., 271} (271 comes from the fact that there are 271 words in the Gettysburg Address)

2) Look up the 10 words in the positions corresponding to the 10 numbers you picked

Pro Tip

Why do we need a random number generator? Well, it turns out that humans are actually pretty bad at truly *randomly* sampling!

To access the random number generator we'll use, please navigate to this link (`https://www.isi-stats.com/isi2nd/ISIapplets2021.html`, for those of you accessing a printed version of this handout) and click on "**Random Number Generator**."

Question 13

(a) Draw 10 random numbers from the set {1, 2, 3, ..., 271} by doing the following:

1) Navigate to the applet above
2) Set "Number of Replications" to 1 and "Numbers per Replication" to 10
3) In the "Number Range," set the "From" value to "1" and the "To" value to "271"
4) Click "Generate"

Write your 10 numbers in the table below:

152	**112**	**257**	**219**	**183**
154	**95**	**145**	**223**	**11**

(b) Compare with your fellow groupmates - did you all get the same 10 numbers, or not?

Solution: Even though you may have *some* numbers that are the same as your groupmates', it's almost guaranteed that your overall set of 10 numbers will be different than any of your groupmates'.

(c) Using the table provided at the back of this Lecture Activity, look up the words corresponding to the 10 numbers you selected in part (a) and record them along with their word lengths in the table below. These 10 words constitute our simple random sample.

Word	Length
we	**2**
of	**2**
here	**4**
fitting	**7**
last	**4**

Word	Length
we	**2**
which	**5**
here	**4**
will	**4**
on	**2**

(d) Fill in the dotplot below with your sampled word lengths.

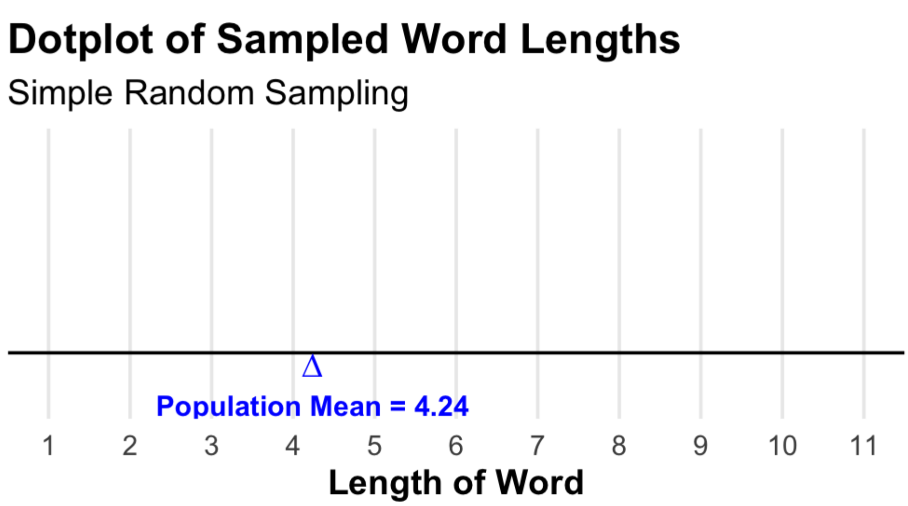


Figure 3: Blank Dotplot

Solution:

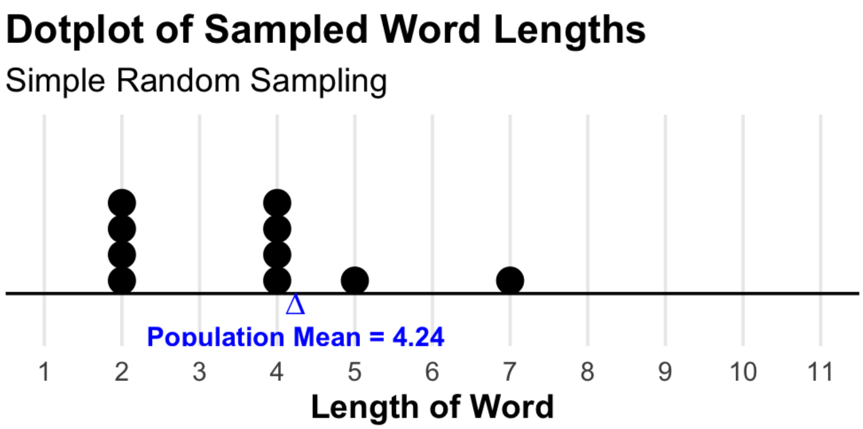


Figure 4: Example answer

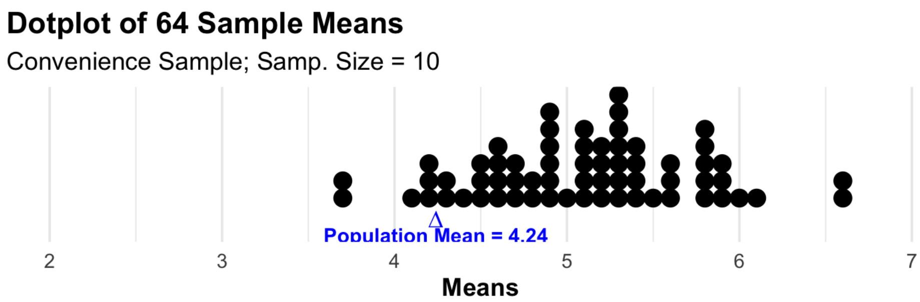
(e) Calculate the average length of your 10 sampled words. Write it here: __**3.6**__ ~~and also on the note card Dr. Ethan gave to you, and bring your notecard to Dr. Ethan (so he can collect everyone's notecard and make an overall dotplot of the class's results!)~~ **correction: we had everyone come up to the front of the room and draw their dot on a dotplot on the whiteboard**.

After most people submit their data, we'll explore the following questions together:

- Approximately ___________ of the class had samples with average lengths larger than 4.24.
- Approximately ___________ of the class had samples with average lengths less than 4.24.
- Approximately ___________ of the class had samples with average lengths exactly equal to 4.24.

While we wait, here is a rough idea of what our final dotplot will look like based on simulated data:

We didn't get a chance to complete this as a class.



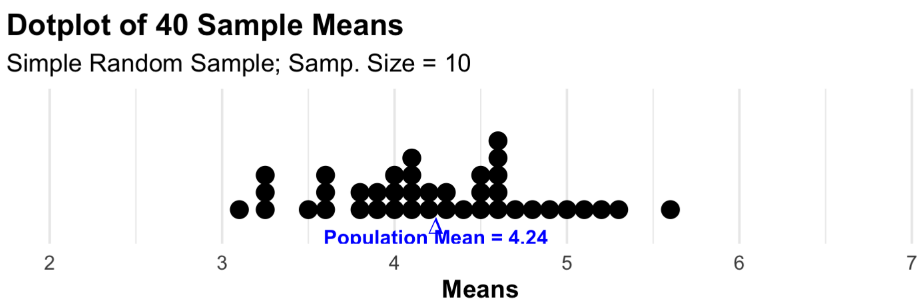


Figure 5: A comparison of dotplots of sample average word lengths. Top plot: convenience sampling. Bottom plot: simple random sampling.

Hopefully this makes it a bit clearer what we mean by biased and unbiased sampling methods - notice how convenience sampling typically results in samples of uncharacteristically long words, whereas simple random sampling typically results in samples that are more "balanced" about the true population average value. This is precisely what we meant when we identified our convenience sampling method as overestimating the parameter!

A Deeper Dive into Bias

In addition to having a tendency to produce over/underestimates of population parameters, there are some other ways sampling techniques could be biased. In particular, **non-response** bias is a particularly insidious type of bias as it can be difficult to detect at first glance. Picture this: a company wants to determine the general office climate. To that end, they take care to select a simple random sample of 500 employees to which they send a survey asking general questions ("on a scale of 1-10, how hostile do you find the office?" or "do you feel welcomed when you are in the office?" etc.). They get back 400 responses (a pretty hefty sample size!) and find the results to be generally favorable: people feel very welcomed, and on equal footing with one another, and nobody reports any feelings of discrimination. Well, it turns out that the 100 people who did not fill out the survey accounted for 80% of the underrepresented minority employees hired by the company (female- and gender-non-conforming employees, non-white employees, employees belonging to the LGBTQIA+ community, etc.). **Do we think the results of the survey are truly unbiased, even though a simple random sample was taken?**

Probably not! That's because we've encountered non-response bias. We can define "non-response bias" to be the case where certain units are included in the sample but, for one reason or another, do not end up having data recorded on them. This is also tied to the notion of **missing data** - it's always very important to ask yourself why is the missing data missing? If there is a pattern underlying the missingness, then we need to be very careful about generalizing our results to the population.

I'd recommend you familiarize yourself with, at the very least, the definition/notion of non-response bias and ensure you can recognize it in the context of a problem!

I'd also like to reiterate the fact that we cannot overcome the effects of bias (including non-response bias) by simply increasing the sample size. For instance, in our office climate example above - simply sending the survey to more people (e.g. in other departments/locations) won't necessarily fix the non-response bias issues. Rather, to truly overcome the effects of bias, we need to carefully examine and modify our sampling technique, not just our sample size.

(Optional) Non-Response vs. Systematic Bias

Non-response bias can seem similar to another type of bias, called systematic bias. Indeed, both non-response and systematic bias result in missing data; however, the mechanisms by which the data is missing differs. In non-response bias, an attempt is made to reach every element of the population (or, at least, the sampling frame) but not every unit responds. In systematic bias, large swaths of the population (or, at least, the sampling frame) aren't even included in the sample! For instance, in the office climate example, systematic bias would be induced if the company just didn't send the survey to any of their URM (under-represented minority) employees. Something for you to think about: is it possible to have non-response bias in a simple random sample? What about systematic bias?

Word Positions, Words, and Word Lengths from the Gettysburg Address; Page 1

Position	Word	Length
1	Four	4
2	score	5
3	and	3
4	seven	5
5	years	5
6	ago	3
7	our	3
8	fathers	7
9	brought	7
10	forth	5
11	on	2
12	this	4
13	continent	9
14	a	1
15	new	3
16	nation	6
17	conceived	9
18	in	2
19	Liberty	7
20	and	3
21	dedicated	9
22	to	2
23	the	3
24	proposition	11
25	that	4
26	all	3
27	men	3
28	are	3

Position	Word	Length
29	created	7
30	equal	5
31	Now	3
32	we	2
33	are	3
34	engaged	7
35	in	2
36	a	1
37	great	5
38	civil	5
39	war	3
40	testing	7
41	whether	7
42	that	4
43	nation	6
44	or	2
45	any	3
46	nation	6
47	so	2
48	conceived	9
49	and	3
50	so	2
51	dedicated	9
52	can	3
53	long	4
54	endure	6
55	We	2
56	are	3

Position	Word	Length
57	met	3
58	on	2
59	a	1
60	great	5
61	battlefield	11
62	of	2
63	that	4
64	war	3
65	We	2
66	have	4
67	come	4
68	to	2
69	dedicate	8
70	a	1
71	portion	7
72	of	2
73	that	4
74	field	5
75	as	2
76	a	1
77	final	5
78	resting	7
79	place	5
80	for	3
81	those	5
82	who	3
83	here	4
84	gave	4

Word Positions, Words, and Word Lengths from the Gettysburg Address; Page 2

Position	Word	Length
84	gave	4
85	their	5
86	lives	5
87	that	4
88	that	4
89	nation	6
90	might	5
91	live	4
92	It	2
93	is	2
94	altogether	10
95	fitting	7
96	and	3
97	proper	6
98	that	4
99	we	2
100	should	6
101	do	2
102	this	4
103	But	3
104	in	2
105	a	1
106	larger	6
107	sense	5
108	we	2
109	can	3
110	not	3
111	dedicate	8

Position	Word	Length
112	we	2
113	can	3
114	not	3
115	consecrate	10
116	we	2
117	can	3
118	not	3
119	hallow	6
120	this	4
121	ground	6
122	The	3
123	brave	5
124	men	3
125	living	6
126	and	3
127	dead	4
128	who	3
129	struggled	9
130	here	4
131	have	4
132	consecrated	11
133	it	2
134	far	3
135	above	5
136	our	3
137	poor	4
138	power	5
139	to	2

Position	Word	Length
140	add	3
141	or	2
142	detract	7
143	The	3
144	world	5
145	will	4
146	little	6
147	note	4
148	nor	3
149	long	4
150	remember	8
151	what	4
152	we	2
153	say	3
154	here	4
155	but	3
156	it	2
157	can	3
158	never	5
159	forget	6
160	what	4
161	they	4
162	did	3
163	here	4
164	It	2
165	is	2
166	for	3
167	us	2

Word Positions, Words, and Word Lengths from the Gettysburg Address; Page 3

Position	Word	Length
168	the	3
169	living	6
170	rather	6
171	to	2
172	be	2
173	dedicated	9
174	here	4
175	to	2
176	the	3
177	unfinished	10
178	work	4
179	which	5
180	they	4
181	who	3
182	fought	6
183	here	4
184	have	4
185	thus	4
186	far	3
187	so	2
188	nobly	5
189	advanced	8
190	It	2
191	is	2
192	rather	6
193	for	3
194	us	2
195	to	2

Position	Word	Length
196	be	2
197	here	4
198	dedicated	9
199	to	2
200	the	3
201	great	5
202	task	4
203	remaining	9
204	before	6
205	us	2
206	that	4
207	from	4
208	these	5
209	honored	7
210	dead	4
211	we	2
212	take	4
213	increased	9
214	devotion	8
215	to	2
216	that	4
217	cause	5
218	for	3
219	which	5
220	they	4
221	gave	4
222	the	3
223	last	4

Position	Word	Length
224	full	4
225	measure	7
226	of	2
227	devotion	8
228	that	4
229	we	2
230	here	4
231	highly	6
232	resolve	7
233	that	4
234	these	5
235	dead	4
236	shall	5
237	not	3
238	have	4
239	died	4
240	in	2
241	vain	4
242	that	4
243	this	4
244	nation	6
245	under	5
246	God	3
247	shall	5
248	have	4
249	a	1
250	new	3
251	birth	5

Word Positions, Words, and Word Lengths from the Gettysburg Address; Page 4

Position	Word	Length
252	of	2
253	freedom	7
254	and	3
255	that	4
256	government	10
257	of	2
258	the	3
259	people	6
260	by	2
261	the	3
262	people	6
263	for	3
264	the	3
265	people	6
266	shall	5
267	not	3
268	perish	6
269	from	4
270	the	3
271	earth	5